Copernicus Land Monitoring Service "Cryosphere and Water"

Framework Service Contract N° 945954(JRC)

QUALITY ASSESSMENT REPORT

GLOBAL LAKE ICE EXTENT (GLOBAL LIE)

COLLECTION 500M GLOBAL

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Organization name of lead contractor for this deliverable: Syke

Book Captain: Kirsikka Heinilä (Syke)

Contributing Authors: Sari Metsämäki (Syke)
Emilia Söderström (Syke)
Sakari Väkevä (Syke)
Jesse Anttila (Syke)
Kari Luojus (FMI)
Gabriele Schwaizer (ENVEO)

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Book captain:	Kirsikka Heinilä	Sign	Date
Approval:	Roselyne Lacaze	Sign	Date
Endorsement:	Nadine Gobron	Sign	Date
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List of Acronyms

ATBD	The Algorithm Theoretical Basis Document
C-GLOPS2	Copernicus Global Land Operations Lot 2
FAR	False Alarm Rate, Comission Error
FMI	Finnish Meteorological Institute
FN	False Negative
FP	False Positive
LIE	Lake Ice Extent
LIE-NH	Lake Ice Extent Northern Hemisphere
MSI	Multi-Spectral imaging Instrument
OLCI	Ocean and Land Colour Instrument
S2	Sentinel-2
S3	Sentinel-3
SLSTR	Sea and Land Surface Temperature Radiometer
Syke	Finnish Environment Institute
TN	True Negative
TP	True Positive

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1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

Copernicus Global Land Operations Lot 2 (C-GLOPS2) is an operation of the Global Land Component, thematic domain Cryosphere and Water. It is allocated as a component of the Land service to ensure the ramp-up and the operational production of 'Cryosphere and Water' related bio-geophysical parameters. Global Lake Ice Extent Northern Hemisphere (Global LIE) is one of the Copernicus Global Land Service products. The Global LIE product is an ice classifier for freshwater bodies and is provided on a daily basis.

The Global LIE classifies, on pixel basis, inland/freshwater bodies as (1) Fully snow-covered ice, (2) Partially snow-covered or snow-free ice, (3) Open water and (4) Cloud. A certain class is associated to a pixel following the rule of highest probability. This probability is also given per pixel to describe the uncertainty of the classification. The gridded data product covers the globe [180°W – 180°E, 60°S – 80°N] in the WGS 84 geographic coordinate system with a 0.005° resolution (~500 m at the equator, decreasing with latitude). Some lakes in the warmest regions of this area are excluded, as they do not experience ice cover throughout the year.

This document presents the results of the quality assessment relevant to the Global LIE product, based on validation conducted for the LIE-NH product.

1.2 SCOPE AND OBJECTIVES

One of the main objectives of the Copernicus programme is to provide information on several environmental variables to the scientific community, as well as other stakeholders, including policy makers. The products are operationally generated and delivered freely through the Copernicus CLMS portal (https://land.copernicus.eu/en/map-viewer) and the Copernicus Data Space Ecosystem (https://dataspace.copernicus.eu/) in near real-time, as well as offline.The purpose of this document is to provide users with the necessary information to evaluate whether the Global LIE product, in terms of its performance characteristics, is suitable for their intended applications.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements, and the expected performance
- Chapter 3 describes the methodology for quality assessment, the metrics, and the criteria of evaluation

- Chapter 4 presents the results of the analyses
- Chapter 5 summarizes the main conclusions of the study
- Chapter 6 makes recommendations based on the results

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1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: GCOS – 245: The 2022 GCOS ECVs Requirements (GCOS 245), WMO, 2022

AD4: Copernicus Global Land Component Lots 1, 2 & 3 Operation of the Global Land Component (Framework Service Contract N° 199494, 199496, 199497, 941119): Service Specifications (CGLOPS-SSD). Issue 1.00. 24th November 2016.

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.
CGLOPS2_QAR_ LIE-NH_500m-V1.3	Quality Assessment Report of the LIE-NH 500m product
CGLOPS2_ATBD_LIE-Global-500m-V2.0	Algorithm Theoretical Basis Document of the Global LIE 500m product

1.4.3 Output

Document ID	Descriptor
CGLOPS2_QAR_ LIE-Global_500m-V2.0	Report describing the results of the scientific quality assessment of the LIE-NH 500m product, which uses the same methodology, with small changes, as the Global LIE product at the Northern Hemisphere scale

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2 REVIEW OF USER REQUIREMENTS

According to the applicable document [AD2], the user's requirements relevant for Global LIE 500m are:

- **Definition**: Global Lake Ice Extent (Global LIE)

- **Geometric properties**:
  - The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view.

Pixel coordinate reference is the upper left corner of the pixel.

- **Geographical coverage**:
  - Projection: geographical latitude, longitude
  - Geodetical datum: WGS84, EPSG:4326
  - Pixel size: 0.005° (ca 500 m, depending on latitude)
  - Coordinate position: pixel centre
    - Window coordinates:
      - Upper left: 180ºW, 80ºN
      - Lower right: 180ºE, 60ºS

- **Accuracy requirements:**

  Target requirements for lake ice cover based on Global Climate Observing System (GCOS) (AD3) are following:

  - Horizontal resolution: 1000 m (Breakthrough[1]: 100 m)
  - Temporal resolution: 3–7 days (Breakthrough: 1 day)
  - Required Measurement Uncertainty (2-sigma): ≤10%

User requirements related to Global LIE products are described by the GCOS – 245: The 2022 GCOS ECVs Requirements (GCOS 245) (AD3) and in User Requirements of CryoLand Snow and Land Ice services - Cryoland (Malnes et al. 2015).

The main user groups with an interest in these satellite products are the climate and hydrological modelling communities, and weather forecasting authorities. The following paragraphs will summarize thematic, temporal and geometrical aspects concerning the Global LIE product.

Thematic aspects

The Global LIE product classifies a section of the freshwater body as either Fully snow-covered ice, Partially snow-covered or snow-free ice, Open water or Cloud. The extent is usually defined in terms of area (in square kilometres). The targeted minimum measurement uncertainty is 10%, while the

goal is 1% according to AD3. The thematic overall accuracy for Global LIE reaches the goal being 99%, with a 1% ice commission error and a 1% ice omission error. In the validation process, two different thresholds were utilized in the generation of 500 m comparison dataset by aggregating the Sentinel 2 -based 20 m reference data into 500 m resolution binary (water/ice) information: (1) a 500 m pixel is classified as ice-covered if the proportion of 20 m ice pixels is >50%, and (2) 500 m pixel is classified as ice-covered if the proportion of 20 m ice pixels is >70%. The validation was conducted for the Northern Hemisphere, but it provides a good representation of the overall accuracy of the global product, as most seasonally ice-covered lakes are located in this region. The validation included more than 10 million product pixels, covering a wide range of lake types across diverse climatic and illumination conditions.

Geometrical aspects

Monitoring the ice on lakes on a global scale, with a daily coverage, requires the use of moderate resolution imagery. Sentinel-3 SLSTR (A/B) data provide good daily coverage over the globe, especially over areas where the freezing of lakes is most prominent. The current spatial resolution for global and basin-scale applications for the parameter Global LIE is 500 m for optical data, with a location accuracy of one pixel.

Temporal aspects

The Global LIE 500 m is a daily product. Spatially and temporally consistent data sets are important for climate modelling and numerical weather prediction. Climate modellers need data sets over large areas (continental scale) and for long time periods (decades). Aggregated (gridded) data sets are preferred. The data sets do not need to be very frequent in time (monthly information is sufficient according to Climate modellers). For other than climatic use, Malnes et al. (2015) summarize that the majority of users request a one day temporal resolution. When often bridging the wishful thinking from users with what is actually possible to achieve with current and near future satellite sensors, it should be realistic to obtain a temporal resolution of one week, due to frequent cloud cover at high latitudes where lake ice mostly occurs. Hence, there is a need to also provide a level of refinement to the ice products, where temporal interpolations provide the desired temporal regularity.

Near real-time data:

Climate modellers have little need for real-time data: annual updates (mostly concerning ice break-up) are sufficient.

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In weather prediction there is in principle a need for near real time data, or at least data for the last day. The most urgent need for real time data seems to arise from the hydrological community, for whom the real time data is a pre-requisite for flood management.

Other aspects

Satellite data on freezing and break-up dates of lake ice is also of interest for all approached users, particularly for the validation of models. These data types can also be of interest for environmental assessment and for biological monitoring.

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3 QUALITY ASSESSMENT METHOD

3.1 DATA AND OVERVALL PROCEDURE

The Global Lake Ice Extent product (Global LIE) is generated daily from medium resolution optical satellite data. The applied *ICEmod* algorithm [CGLOPS2_ATBD_LIE-Global-500m_V2.0] classifies each 0.005° (referred here as 500m) resolution lake pixel as (1) Fully snow-covered ice, (2) Partially snow-covered or snow-free ice, 3) Open water or (4) Cloud. The validation was conducted for LIE-NH, not for Global LIE, since the algorithm and the relevant lake coverage (iced-lakes) of Global LIE are so similar to the LIE-NH, that the producers believe the validation of LIE-NH can be used as is to get enough information on the quality of Global LIE. In short, validating Global LIE, according to the producers, is not necessary, and would be rather a duplication of the work done for LIE-NH. Thus, the results do not differ from those reported in the QAR for LIE-NH (see CGLOPS2_QAR_LIE-NH-500m-V1.2). However, it is important to notice that since the validation was performed for LIE-NH, no metrics for the classes (1) Fully snow-covered ice and (2) Partially snow-covered ice are presented, as these classes are not implemented separately for LIE-NH, but are included under one “Ice” class.

The validation was carried out by comparing 20 m high-resolution Lake Ice Extent (LIE) products with the LIE-NH 500m products, which also use the same ICEmod methodology (Heinilä et al. 2021) as the Global LIE. The reference LIE maps were generated from dataset of 0.0002° (referred here as 20 m) resolution Sentinel-2 (S2) Multi-Spectral imaging Instrument (MSI) satellite images. The validation was conducted for the years 2020–2022, covering an extensive selection of different types of lakes (with respect to water quality, size, and geographical location) and various ice conditions. All selected lakes undergo seasonal ice cover. Overall, the S2 MSI dataset included thousands of images from 45 different lakes, 15 per each continent (Asia, Europe, and North America) (Figure 1). All cloudless or nearly cloudless S2 MSI images covering these lakes for the entire period from 1 January 2020 to 31 May 2022 were included. This way it was possible to assess the product performance and accuracy quite reliably. However, the unavailability of LIE-NH product at the time of the polar darkness for the northernmost lakes reduces the validation data during the late-autumn and mid-winter.

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Figure 1. Map representing the spatial coverage of selected lakes. Bounding boxes illustrate the footprint of the reference scenes.

The S2 MSI images were first pre-processed by resampling all bands in 0.0002º (~20-meter) resolution and reprojecting the image to be equal with the LIE-NH product. By using extensive and well-prepared training data, including over 38 million classified ice or water pixels, a Decision Tree model was created to pre-classify the reference images into the lake ice classes. Afterwards, all S2-based products were manually quality checked and corrected when necessary. Figure 2 illustrates the automatically classified LIE -product for Rybinsk reservoir in Russia on 19 April 2021. The complexity of ice cover is visible in S2 MSI RGB image (Figure 2, left) and the decision on whether the pixel is ice or not is difficult to make on a visual basis. Since the algorithm utilizes several visible, near- and shortwave infrared bands, in complex cases the best decision was provided automatically by the algorithm. After quality check, and some manual corrections, 1722 S2 MSI -based LIE products served as a reference data including over 10 million (10 582 778) cloud-free 500-meter LIE-NH lake pixels for pixel-by-pixel intercomparison. The utilized reference data is described below:

- 44.5% of the data represent lakes that had both ice-covered and open-water pixels in the same observing time
- 32.5% of the data represent lakes that had only ice pixels in the same observing time
- 23.0% of the data represent lakes that had only water pixels in the same observing time

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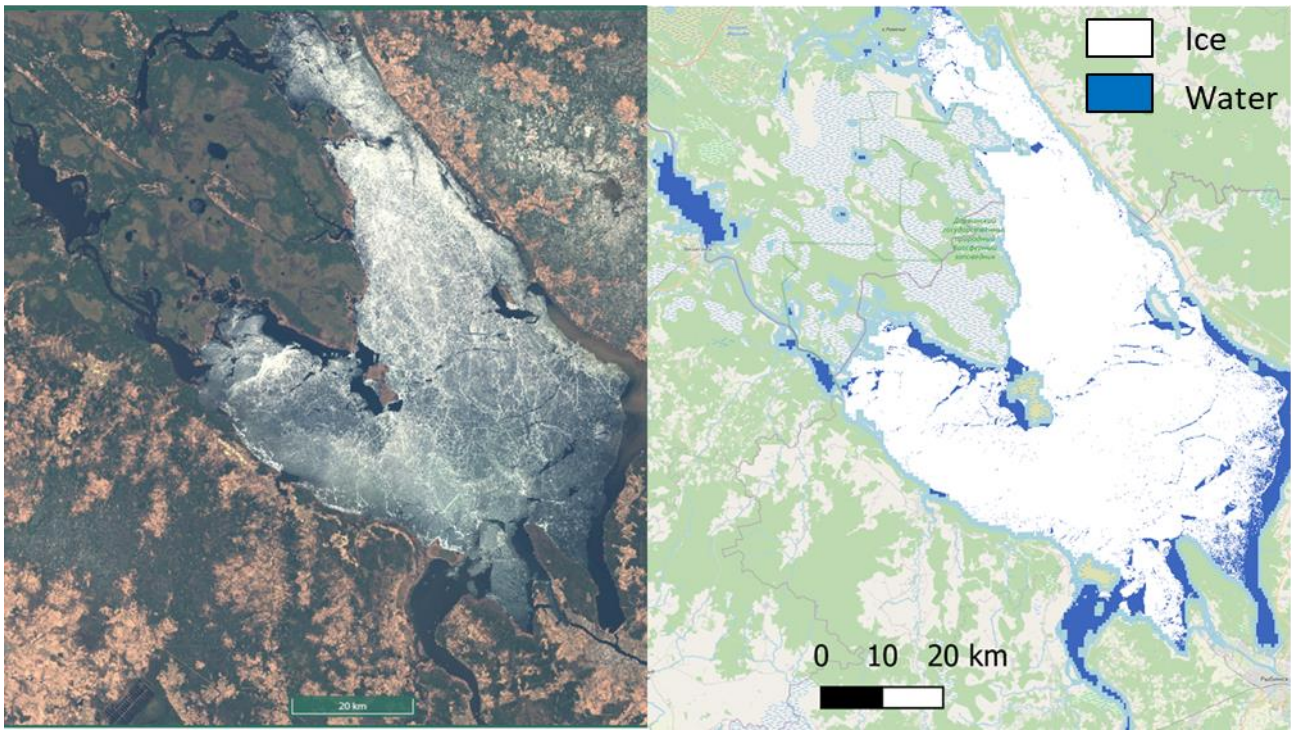


Figure 2. Rybinsk reservoir, Europe (58°22′N 38°26′E), on 19 April 2021. Left: S2 MSI RGB image. Right: Automatically generated 20-meter resolution S2 MSI-based LIE product with buffered shoreline to be equal with the LIE-NH 500m product.

For pixel-by-pixel comparison in 500 m resolution of LIE-NH product, the quality checked S2-based LIE products were generated to 500 m binary ice/water maps first by calculating the ice-cover fraction from 20 m pixels within each 500 m grid cell, and then by converting each pixel's fraction to binary information. In the binary conversion two alternative thresholds were applied: either the 500 m reference pixel is 'ice' if its ice cover fraction > 50%, or alternatively if its ice cover fraction > 70%. The remaining pixels are 'no-ice'. Two different threshold values were employed in generating 500 m reference pixels to ensure a thorough evaluation of our ice product's classification efficiency under diverse conditions. The thresholds 50% and 70% were chosen to represent different scenarios and use cases, the 50% reflecting a broader overview suitable for applications where the general presence of ice is of interest, while the 70% threshold provides a more stringent criterion relevant for applications requiring the accurate identification of significant ice cover. The results are also presented for both thresholds. The reference LIE products were compared pixel by pixel to LIE-NH products. All cloud, land, sea, and no-data pixels in either the LIE-NH product or the reference lake ice extent product were excluded from the analysis. The data pairs were formed so that the value of a product pixel and the underlying reference pixel are from the same day. The time difference between the S2 MSI images and S3 SLSTR images utilized in LIE products is at maximum a few hours, indicating that the ice conditions should be the same.

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Finally, the agreement between the original S3 SLSTR-based and the high resolution -based binary (ice/no-ice) LIE-products was investigated by forming a contingency matrix indicating the numbers for True Positives (TP, true ice correctly classified as ice), True Negatives (TN, true no-ice correctly classified as no-ice), False Positives (FP, true no-ice incorrectly classified as ice) and False Negatives (FN, true ice incorrectly classified as no-ice). From this matrix, commonly established statistical metrics are calculated to assess the accuracy of the Copernicus Global land LIE-NH operative products (

Table 1). Additionally, the omission and commission errors are provided. The concept for the product validation with high resolution optical satellite data is shown in Figure 3.

Table 1. Statistical metrics used to describe the thematic accuracy of the Sentinel-3 SLSTR -based LIE-NH product validated with high resolution Sentinel-2 MSI data.

Measure	Equation	Description
Recall	TP / (TP+FN)	The ratio of correctly classified ice pixels to all true ice pixels. The product's ability to find ice.
Overall accuracy (Hit rate)	(TP+TN) / (TP+FP+TN+FN)	A metric to identify the ratio of correctly classified (reference) pixels to the total number of (reference) pixels.
Omission error	FN / (TP + FN)	Percentage of true ice pixels falsely classified as open water out of all true ice pixels
Commission error, False Alarm Rate	FP / (FP + TN)	Percentage of true open water pixels falsely classified as ice out of all true open water pixels
Precision	TP / (TP+FP)	The ratio of correctly classified ice pixels to all pixels classified as ice. Describes the certainty of ice identifications.
F-score	2 * TP / (2 * TP + FP + FN)	The weighted harmonic mean of the precision and recall

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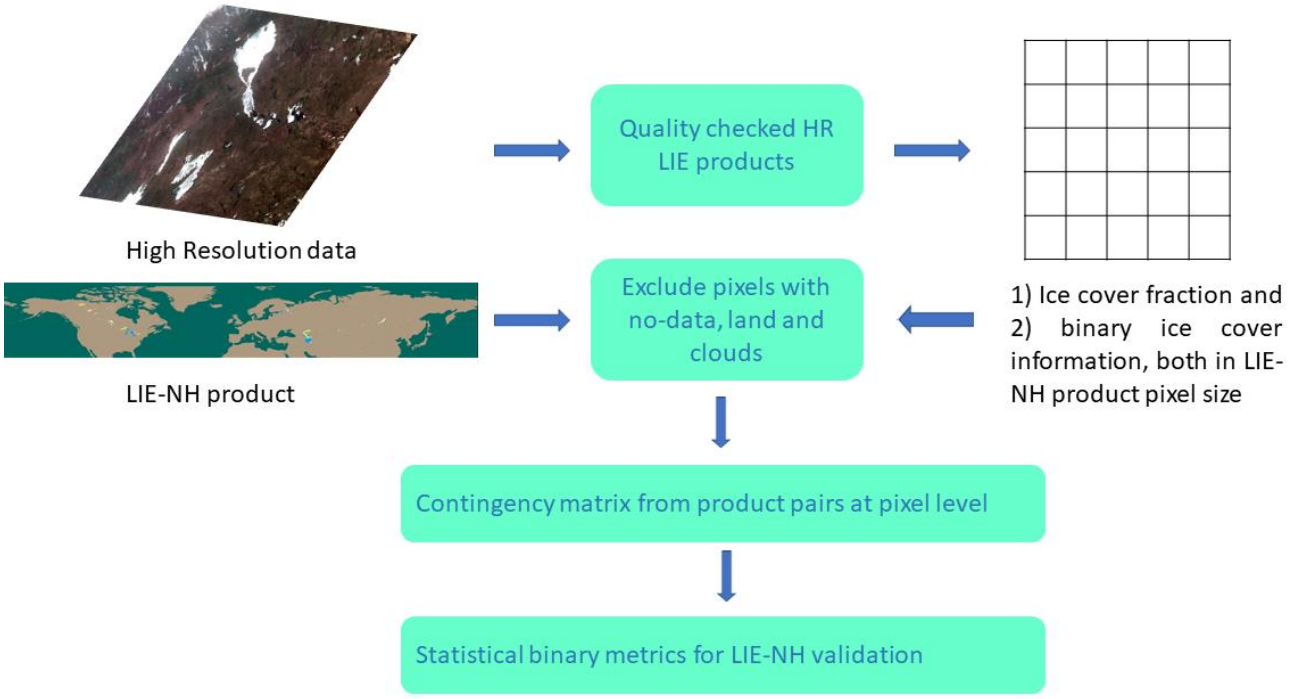


Figure 3. The concept for the product validation with high resolution optical satellite data

The validation of cloud class is hampered by the fact that there is no valid product that detects clouds from S3 SLSTR data over northern lakes well enough to be used as a "truth". Figure 4 present cloudy conditions over Lake Balkhash, Kazakhstan, on 5 January 2022. Some of the clouds are not detected by any of the cloud layers of the S3 SLSTR data package. Instead, LIE-NH recognizes clouds very well (Figure 4 Bottom). Additionally, the manual cloud mask production is complicated due to the difficulties in deciding on whether a pixel of the reference data should be classified as cloud or not. Often, the human eye (as well as the *ICEmod* algorithm) can distinguish ice conditions beneath the "cloud", e.g., fog over ice, mist over water. Thus, the ability of *ICEmod* to identify clouds was performed by visually comparing dozens of LIE-NH products with S3 Ocean and Land Colour Instrument (OLCI)/SLSTR images.

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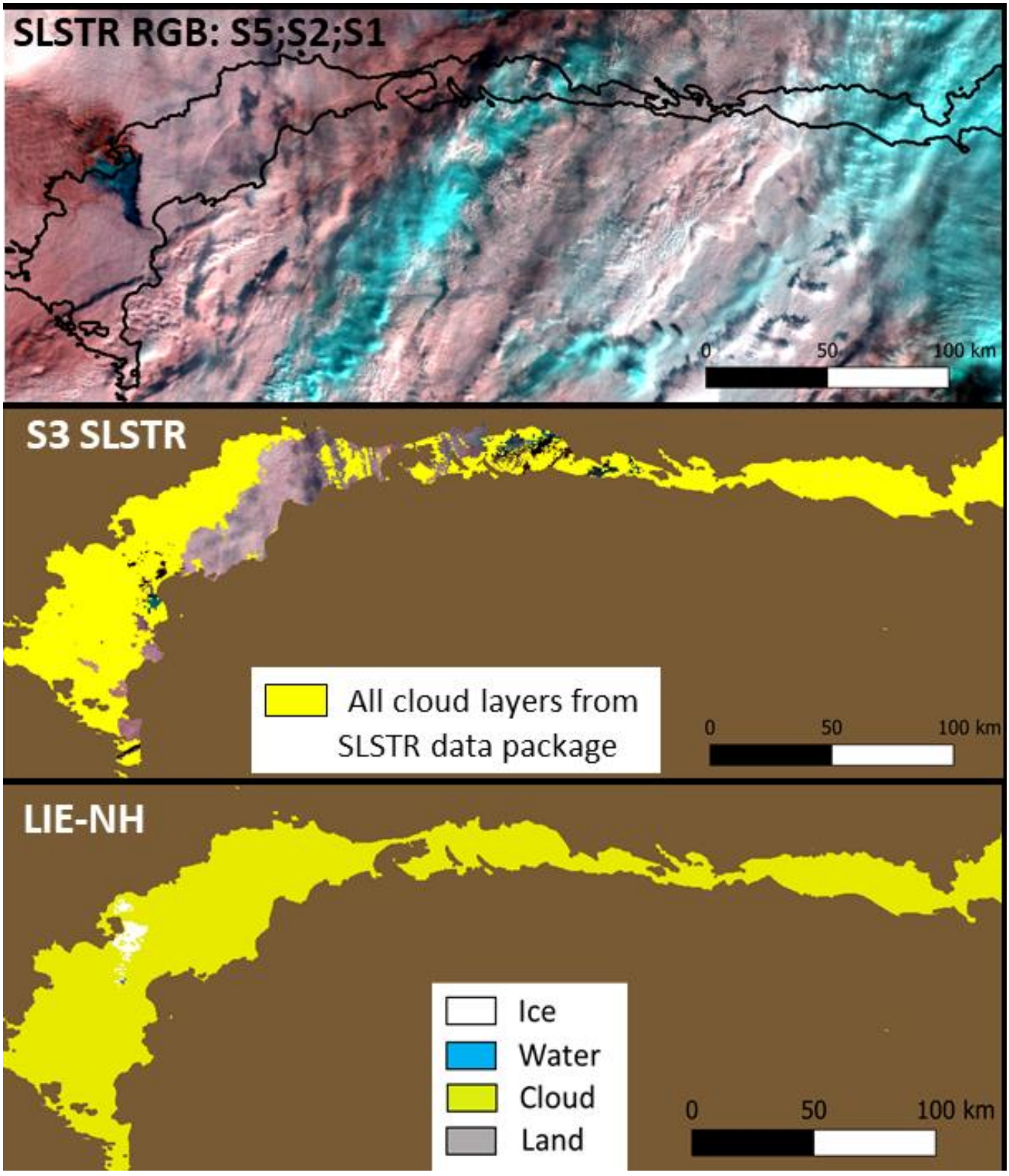


Figure 4. Lake Balkhash, Asia (46°10′N 74°20′E), on 5 January 2022. Top: S3 SLSTR RGB (B5;B2;B1) image where almost the entire area is seemingly cloud-covered. Middle: S3 SLSTR RGB image with all cloud layers (included in the S3 SLSTR data package) added on and shown as a yellow. Bottom: LIE-NH product.

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4 RESULTS

4.1 VALIDATION RESULTS FOR THE WHOLE DATASET

LIE-NH product is validated against 1722 high resolution S2 MSI -based LIE products. The validation dataset includes over 10 million cloud-free 500-meter LIE-NH lake pixels for pixel-by-pixel intercomparison. The southernmost site is at latitude 30.52°N while the northernmost site is at a latitude of 72.76°N. The time frame of the selected images is from 15 January 2020 to 31 May 2022, including images from all seasons. The validation sites are spread across a wide range of latitudes, including different types of lakes. Figure 5 presents validation cases for different seasons in Uvs Lake, which is a shallow and a highly saline lake located at 50°18′N 92°42′E in Asia. Table 2 presents the contingency matrix for all validated LIE-NH pixels, including results using both alternative thresholds for fractional to-binary conversion (see 3.1). The corresponding binary metrics are presented in Table 3.

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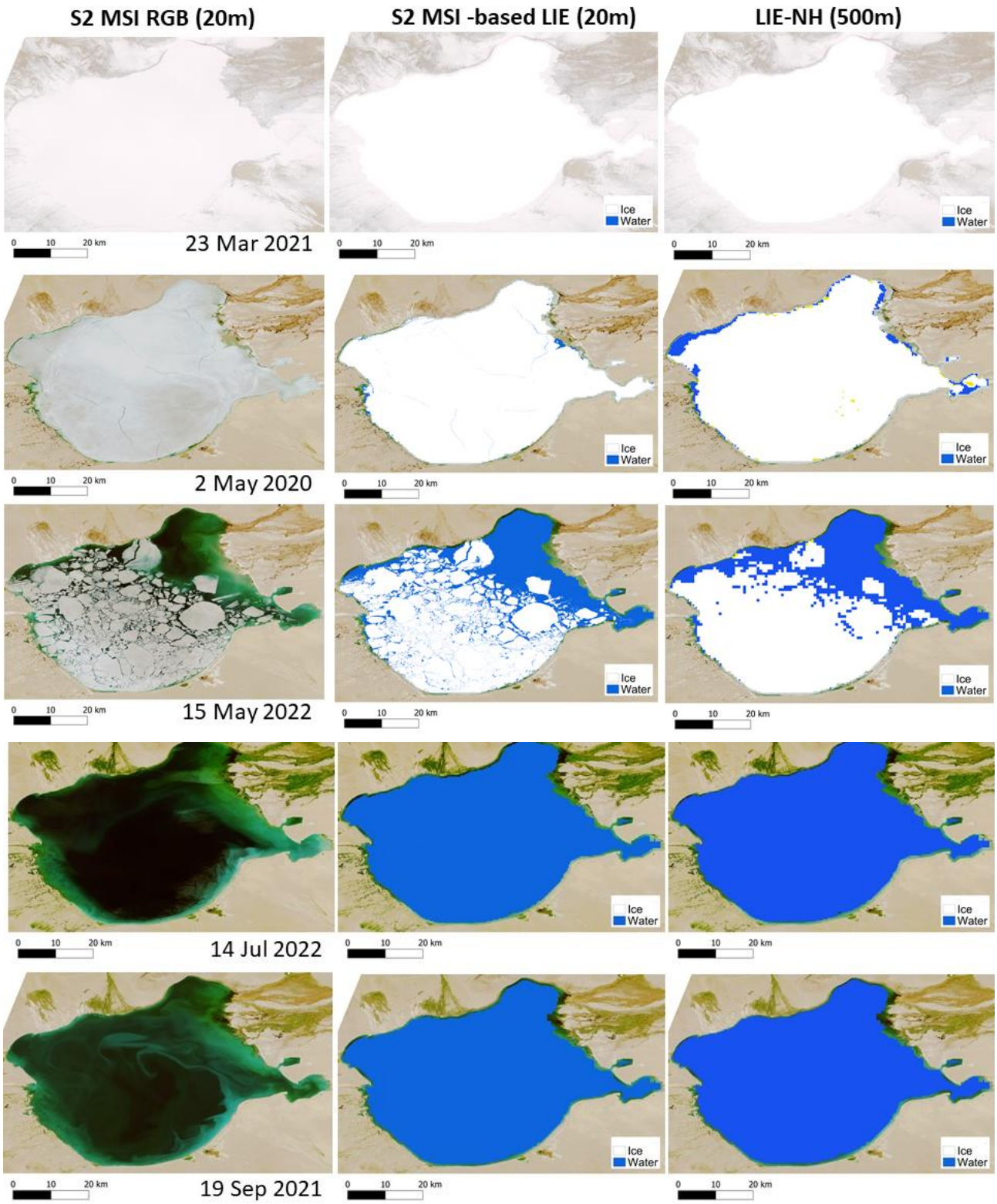


Figure 5. Validation cases for different conditions in Uvs Lake, Asia (50°18′N 92°42′E). Left pane: S2 MSI-RGB images. Middle pane: 20 m resolution S2 MSI -based LIE product. Right pane: 500 m resolution S3 SLSTR -based LIE-NH product.

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Table 2. Contingency matrix for all validated LIE-NH 500m pixels. Values in normal font represent the cases where 500 m reference pixel is classified as ice if it includes 50% or more 20 m resolution ice pixels. Values in bold font represent the cases where 500 m reference pixel is classified as ice if it includes 70% or more 20 m resolution ice pixels.

	LIE-NH 500m		
S2 MSI -based LIE 500m	No ice	Ice	Number of cases
No Ice	3831440 **3817805** (TN)	23488 **14493** (FP)	3854928 **3883052**
Ice	75697 **89332** (FN)	6651153 **6660148** (TP)	6726850 **6698726**
Number of cases	3907137 **3907137**	6674641 **6674641**	10581778

Table 3. The binary metrics derived from Table 2 (the better ones in bold).

Metrics	Ice-threshold: reference pixel is ice if 50% or more ice-covered	Ice-threshold: reference pixel is ice if 70% or more ice-covered
Recall	98.9%	**99.1%**
Overall accuracy, Hit rate	**99.1%**	99.0%
Omission error	1.1%	**1.0%**
Commission error, FAR	**0.6%**	1.0%
Precision	**99.7%**	99.4%
F-score	**99.3%**	99.2%

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The validation indicates a very good performance of the *ICEmod* method (overall accuracy 99%) even though almost half of the validation dataset (45%) consists of complex cases, i.e., lakes including both ice-covered and open water pixels (Tables 2 and 3). The threshold for converting 20 m -based LIE into 500 m product did not make much difference to the results. Evidently the S3 SLSTR -based LIE products are within required thematic accuracies (uncertainty < 10% according to GCOS – 245: The 2022 GCOS ECVs Requirements (GCOS 245) (AD3)). Omission errors (1.1%) are mainly caused by the water layer on top of the ice, as well as fractional ice with water between ice floes. This is because water absorbs almost all incoming radiation and even a thin water layer on top of dark ice may cause some omission errors, resulting in the pixel being classified as water. If the ice is not dark beneath the water layer, the *ICEmod* algorithm, utilized in the LIE-NH product, can distinguish between ice and water. Figure 5 indicates that the ice has started to thaw from the shores in Uvs lake on 2 May 2020 and most of the ice layer is detected, but a few areas near the shores are misclassified as water. Investigation on False Negatives (FN, true ice incorrectly classified as water) revealed that there were no 500 m reference pixels that had an ice fraction of 100% i.e., all true ice pixels incorrectly classified as water included some 20 m open water pixels. Figure 6 shows the number of FN per month as well as the relative number of FN per month (number of FN / number of all observations). The relative number is a more reliable measure, as polar night and cloud cover limit the observations during the late autumn and mid-winter. Figure 6 shows that the low light conditions in mid-winter (December, January, February) increases the possibility of omission errors.

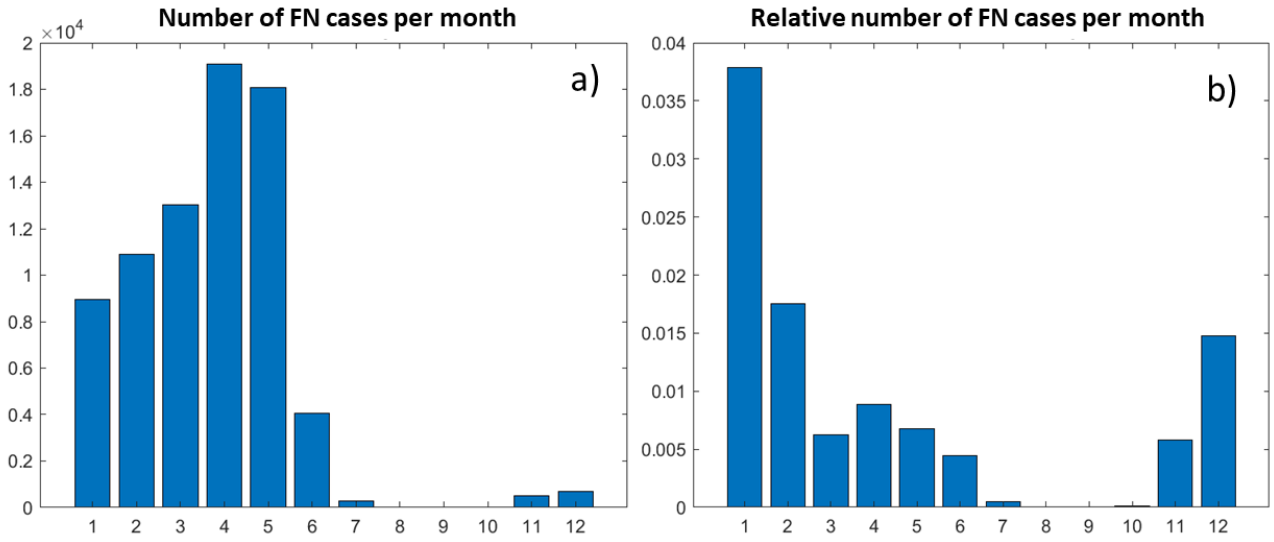


Figure 6. a) Number of False Negatives (FN, true ice incorrectly classified as water) per month. b) Relative number of FN per month i.e., the number of FN divided by the number of total observations per month.

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Commission errors (0.6%) are mainly caused by very thin cirrus or mist above very cold water. Additionally, very heavy turbidity can cause false ice commissions when the water is close to the freezing point and the temperature limit does not apply (see Figure 7 zoomed a) area). The commission error is also affected by the applied threshold for ice classification (50% or 70%) in the reference data. Even rather small pieces of white ice within a 500 m S3 SLSTR pixel may increase the visible reflectance, resulting in the pixel being classified as ice. This is visible in Figure 5 at Uvs Lake on 15 May 2022. Figure 8a shows the number of False Positives (FP, true no-ice incorrectly classified as ice) per month and Figure 8b the number of FP per month when the reference pixel does not include any (20 m) ice pixels. It is evident that the number of FP in Figure 8b decreases significantly, confirming the effect of small ice pieces within the 500 m pixel on the commission error. Figure 8c and Figure 8d show the relative values of Figure 8a and Figure 8b, respectively, illustrating that the mist and low light conditions in late-autumn and mid-winter (November, December and January) increases the possibility of commission errors. Otherwise, temporal comparison between observations is difficult, because apart from the fact that the lakes are located at different latitudes, e.g., the Gulf Stream and the continental climate cause large climatological differences even between areas at the same latitude. The timing of freezing and break-up in different lakes varies significantly. In some cases, the ice season is so short that the freezing and break-up occur within one month. Additionally, polar night and cloud cover limit the observations during the late autumn and mid-winter in most of the lakes.

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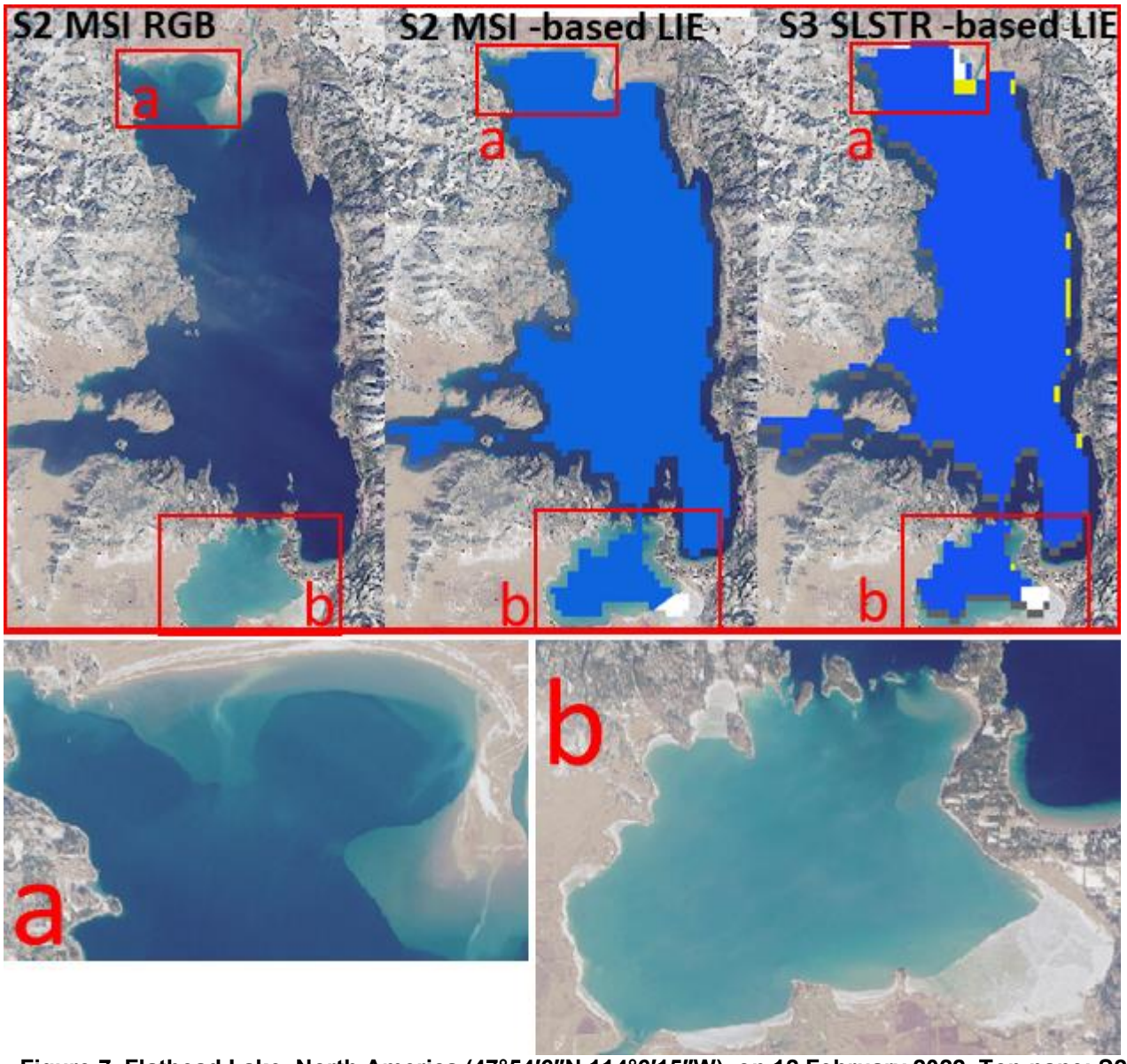


Figure 7. Flathead Lake, North America (47°54′6″N 114°6′15″W), on 12 February 2022. Top pane: S2 MSI-RGB image (left), 20 m resolution S2 MSI -based LIE product (middle) and 500 m resolution S3 SLSTR -based LIE product (right). In LIE products, blue, white and yellow represents water, ice and cloud, respectively. Bottom pane: Zoomed areas from S2 MSI-RGB image.

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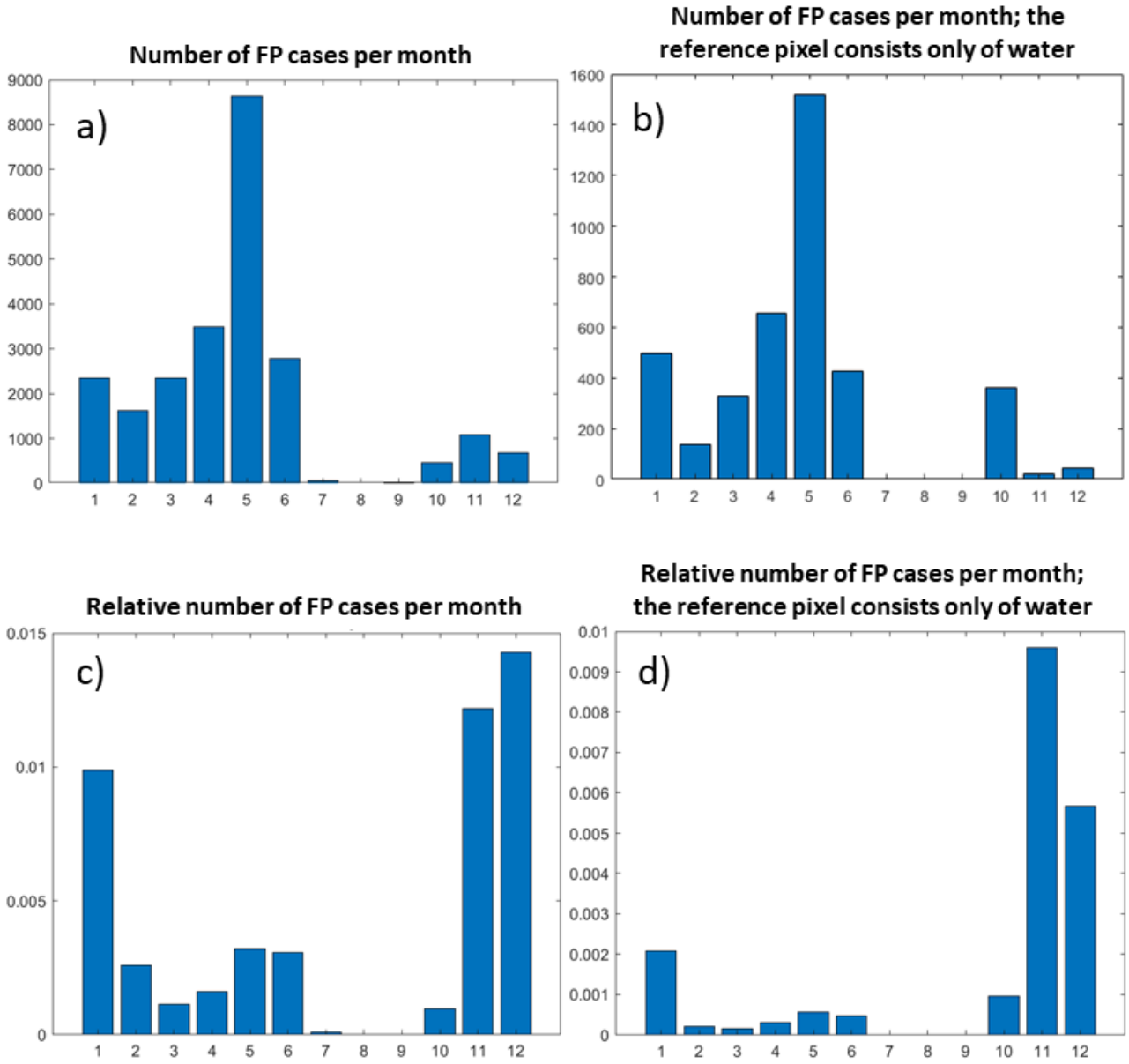


Figure 8. a) Number of False positives (FP, true no-ice incorrectly classified as ice) per month. b) Number of FP per month when the ice fraction in reference data pixel is 0% i.e., does not include any 20 m ice pixel within the 500 m pixel size. c) Relative number of FP per month i.e., the number of FP were divided by the number of total observations per month. d) Relative number of FP per month when the ice fraction in reference data pixel is 0%.

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4.2 VALIDATION RESULTS SEPARATELY FOR LAKES HAVING 1) ONLY WATER PIXELS, 2) ONLY ICE PIXELS, AND 3) BOTH ICE AND OPEN WATER PIXELS

To evaluate the performance of the LIE-NH products in more detail, validation was conducted for three cases separately: 1) reference products with only open-water pixels (23.0% of the dataset); 2) reference products with only ice-covered pixels (32.5% of the dataset); 3) reference products that had both open-water and ice-covered pixels present at the same time (44.5% of the dataset). All these cases cover hundreds of S2 MSI images in different parts of the Northern Hemisphere. In case 1), there are only a very small number of False Positive (FP) ice classifications, resulting in a 0.05% commission error (Tables 4 and 5). These few FP occur due to extremely cold and highly turbid water or fog over very cold water. In case 2) there are some more false negative classifications, resulting in a 0.5% ice omission error (Tables 6 and 7). False classifications in these cases are mainly caused by the presence of a water layer atop dark ice cover, which occurs when the ice has started to thaw but remains unbroken. In case 3) two different thresholds were applied to generate the 500 m LIE products from high resolution reference data (see section 3.1). The results for both cases are presented in Tables 8 and 9. The results indicate that during the freezing and break-up of ice, the product tends to exhibit a higher commission error than omission error. Nevertheless, even during times when the lake is only partly ice-covered, the overall accuracy is over 98%. Figure 9 a) presents the relative number of FP per month for case 3), while Figure 9 b) shows the relative number of FP per month for case 3) when the ice fraction within the reference pixel is 0% i.e., does not include any 20 m ice pixels within the 500 m pixel size. Again, it is clear that ice pieces within the product's 500 m pixel spacing increase the observed reflectance and cause commission errors (i.e., values in Figure 9a are much higher than in Figure 9b). In Figure 9 the effect of low light conditions on the misclassification can be seen as most of the FP occurs during the ice freezing in late autumn and mid-winter (Figure 10). The omission errors during the thawing and freezing of ice are mainly caused by the water layer atop dark ice and again, occur mostly during the low light conditions in mid-winter. For end-users this should not be considered as strong defect, because at that time the ice is already weak (citizens), dark (earth energy budget) and partly open (ecologist, weather modelers).

So, for fully open lakes the commission error is 0.05%, while for fully ice-covered lakes the omission error is 0.5%. This allows us to deduce that the prominent contribution to the general commission error (0.6%) and omission error (1.1%) in Table 3 is by lakes with both ice and open water pixels. According to these results, *ICEmod* method utilized in LIE-NH can provide nearly 100% accuracy for entirely ice-covered lakes and for entirely ice-free lakes.

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Table 4. Contingency matrix for validation dataset where lakes in reference products represent only open water (23.0% of the dataset).

	LIE-NH 500m		
S2 MSI -based LIE 500m	No ice	Ice	Number of cases
No Ice	2432379 (TN)	1162 (FP)	2433541
Ice	0 (FN)	0 (TP)	0
Number of cases	2432379	1162	2433541

Table 5. The binary metrics derived from Table 4. Reference products representing only open water lakes (23.0% of the dataset) are included.

Metrics	Ice-threshold: reference pixel is ice if 50% or more ice-covered
Recall	–
Overall accuracy, Hit rate	99.95%
Omission error	–
Commission error, FAR	0.05%
Precision	–
F-score	–

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Table 6. Contingency matrix for validation dataset where reference products represent only fully ice-covered lakes (32.5% of the dataset).

	LIE-NH 500m		
S2 MSI -based LIE 500m	No ice	Ice	Number of cases
No Ice	0 (TN)	0 (FP)	0
Ice	18320 (FN)	3417199 (TP)	3435519
Number of cases	18320	3417199	3435519

Table 7. The binary metrics derived from Table 6. Reference products representing only fully ice-covered lakes (32.5% of the dataset) are included.

Metrics	Ice-threshold: reference pixel is ice if 50% or more ice-covered
Recall	99.5%
Overall accuracy, Hit rate	99.5%
Omission error	0.5%
Commission error, FAR	–
Precision	100%
F-score	99.7%

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Table 8. Contingency matrix for S3 SLSTR -based LIE-NH 500m pixels where lakes in reference products were partly ice-covered, partly ice-free (44.5% of the dataset) (see section *3.2*). Values in normal font represent the cases where 500 m reference pixel is classified as ice if it includes 50% or more 20 m resolution ice pixels. Values in bold font represent the cases where 500 m reference pixel is classified as ice if it includes 70% or more 20 m resolution ice pixels.

	LIE-NH 500m		
S2 MSI -based LIE 500m	No ice	Ice	Number of cases
No Ice	1399061 **1411213** (TN)	22326 **38253** (FP)	1421387 **1449466**
Ice	57377 **45225** (FN)	3233954 **3218027** (TP)	3291331 **3218027**
Number of cases	1456438 **1456438**	3256280 **3256280**	4712718

Table 9. binary metrics derived from Table 8. Reference products representing only lakes which are partly ice-covered and partly ice-free (44.5% of the dataset) are included.

Metrics	Ice-threshold: reference pixel is ice if 50% or more ice-covered	Ice-threshold: reference pixel is ice if 70% or more ice-covered
Recall	98.3%	**98.6%**
Overall accuracy, Hit rate	**98.3%**	98.2%
Omission error	1.7%	**1.4%**
Commission error, FAR	3.2%	**2.6%**
Precision	**99.3%**	98.8%
F-score	**98.8%**	98.7%

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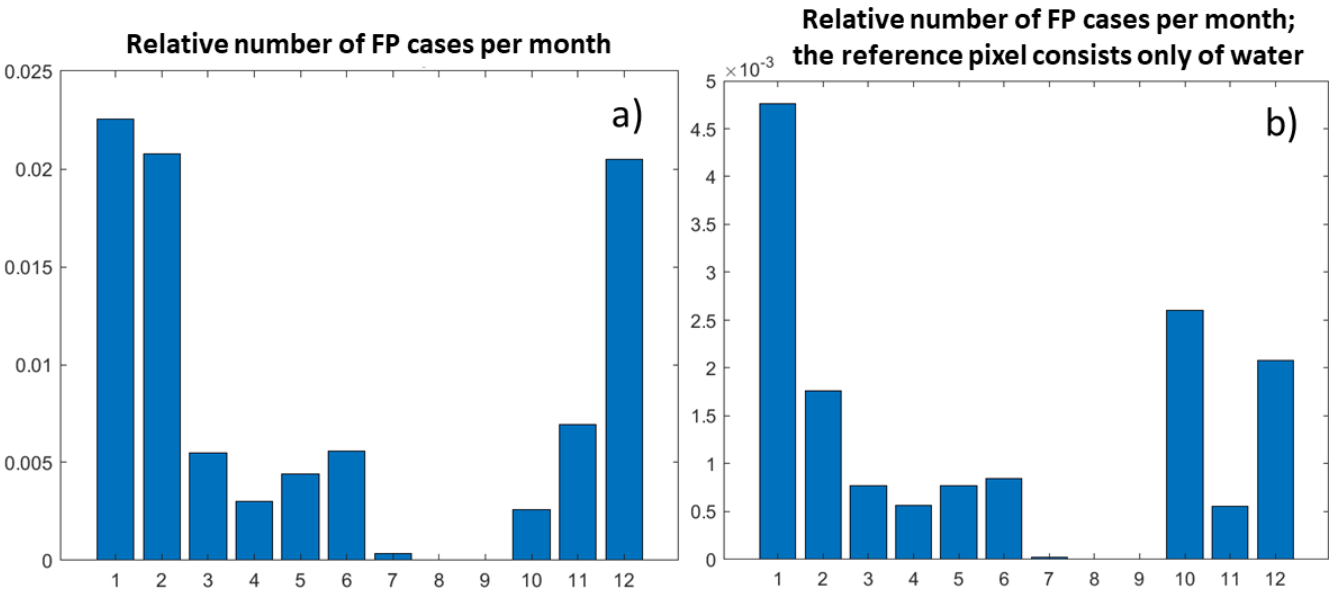


Figure 9. a) Relative number of False Positives (FP, true no-ice incorrectly classified as ice) per month i.e., the number of FP were divided by the number of total observations per month (this data only includes lakes with both water and ice pixels presented at the time of observation). b) Relative number of FP per month when the ice fraction in reference data pixel is 0% i.e., does not include any 20 m ice pixel within the 500 m pixel size.

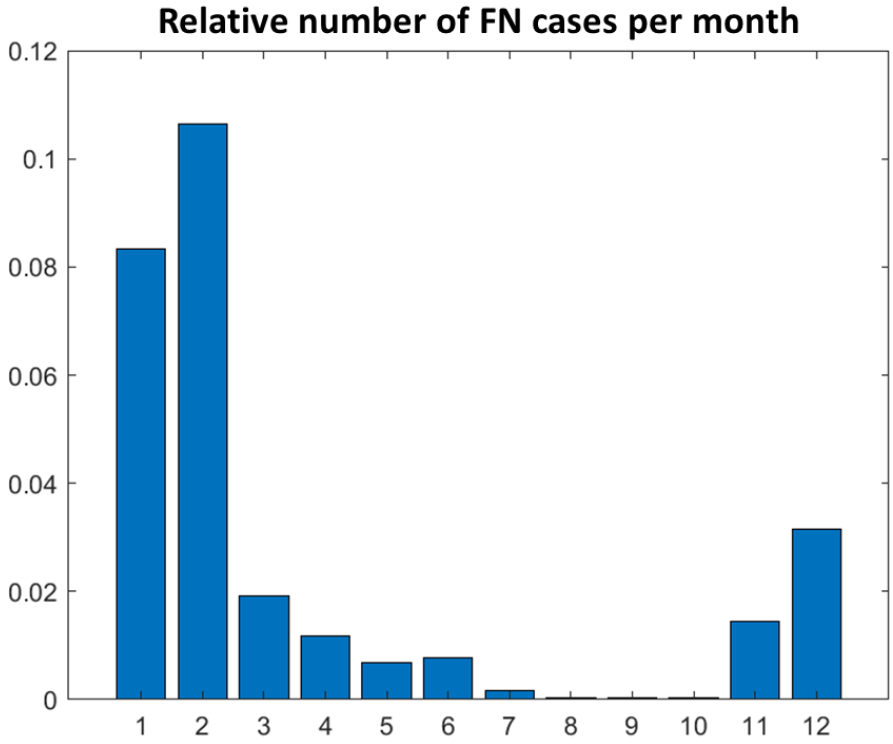


Figure 10. Relative number of False Negatives (FN, true ice incorrectly classified as water) per month i.e., the number of FN divided by the number of total observations per month (this data only includes lakes with both open water and ice pixels presented at the time of observation).

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4.3 VALIDATION OF THE CLOUD CLASS

The validation of cloud pixels was conducted by comparing dozens of LIE-NH products with concurrent S3 OLCI/SLSTR images. Generally, the cloud detection works very well with the *ICEmod* method. The *ICEmod* tends to overestimate clouds rather than to underestimate. Figure 11 shows an example from the Great Lakes on 11 February 2021 where part of Lake Superior and part of Lake Erie is covered by haze, but the ice cover beneath is clearly visible for human eye. In LIE-NH, some of these areas are classified as cloud. The reason for this is the following: *ICEmod* is based on Gaussian distributions and when the distribution indicates optional classes, the cloud class was favoured if it included haze/cloud over ice pixels and haze/cloud over water pixels. During the summertime, despite the haze, open water class is more often chosen, because the use of the thermal data in the algorithm (Figure 12). Overall, the *ICEmod* can discriminate between clouds and different kinds of ice very well (Figures 11 and 13). In late autumn and mid-winter, when the light conditions are low, the incorrectly classified ice pixel may occur rarely. Figure 14 presents Lakes Peipus and Võrtsjärve under cloudy conditions. Even during the low light conditions and cold water temperatures, the *ICEmod* discriminates between cloud and open water areas quite nicely. However, in Võrtsjärve, where the water is very turbid, a few pixels at the edge of cloud are incorrectly classified as ice. The *ICEmod* algorithm provides precision for each pixel, which is lower at some of the misclassified pixels. (Figure 12).

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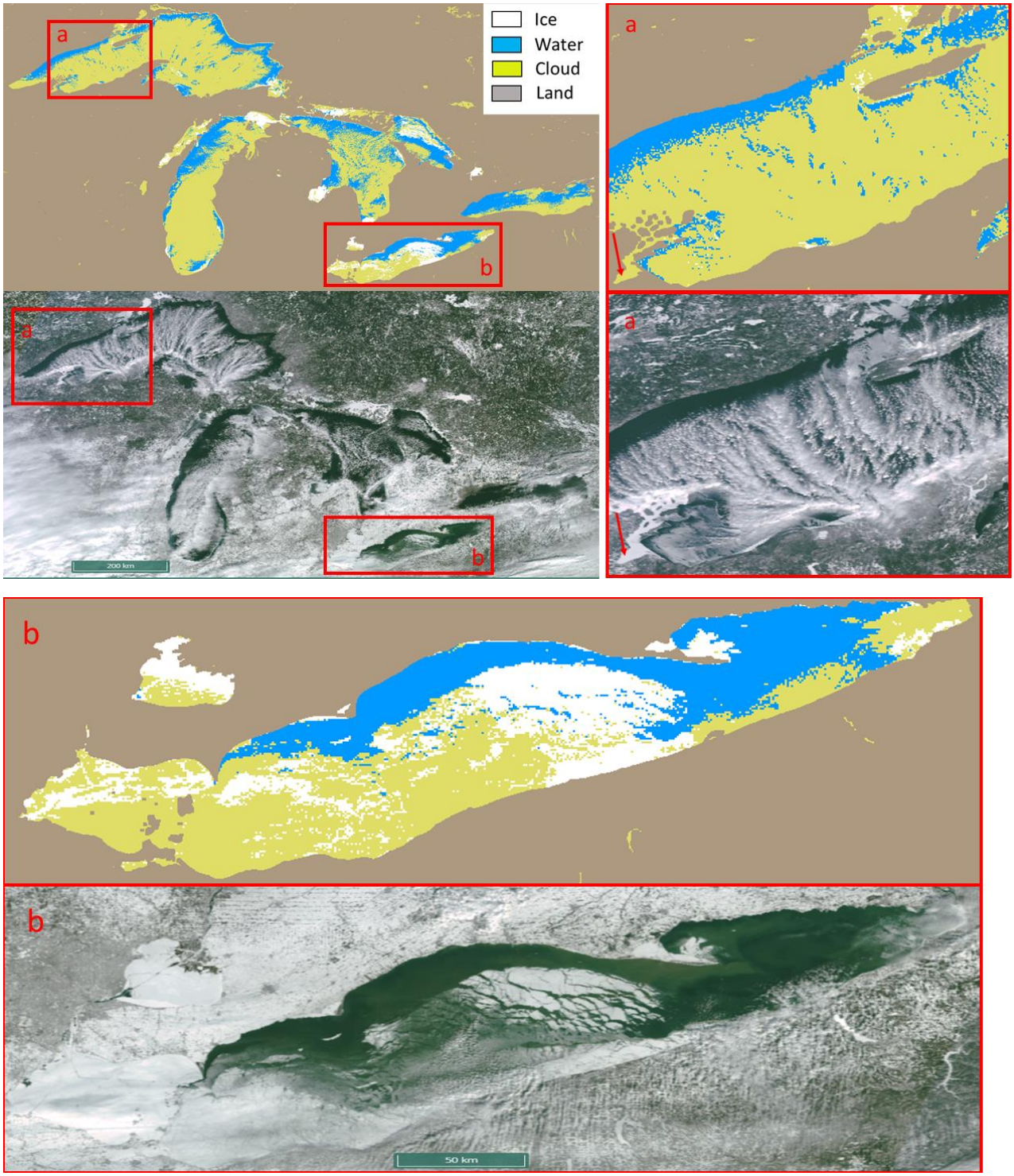


Figure 11. S3 OLCI RGB image and LIE-NH product from the Great Lakes, North America (45°N 84°W), 11 February 2021. a) A comparison for part of Lake Superior indicates that LIE-NH detects clouds well but tends to classify pixel as cloud when only thin cloud layer covers the ice (e.g., the place shown with a red arrow in the top right figure). b) A comparison for Lake Erie also indicates the product's good cloud detection capability but at the same time, sensitivity of the algorithm to detect thin cloud layers especially over ice.

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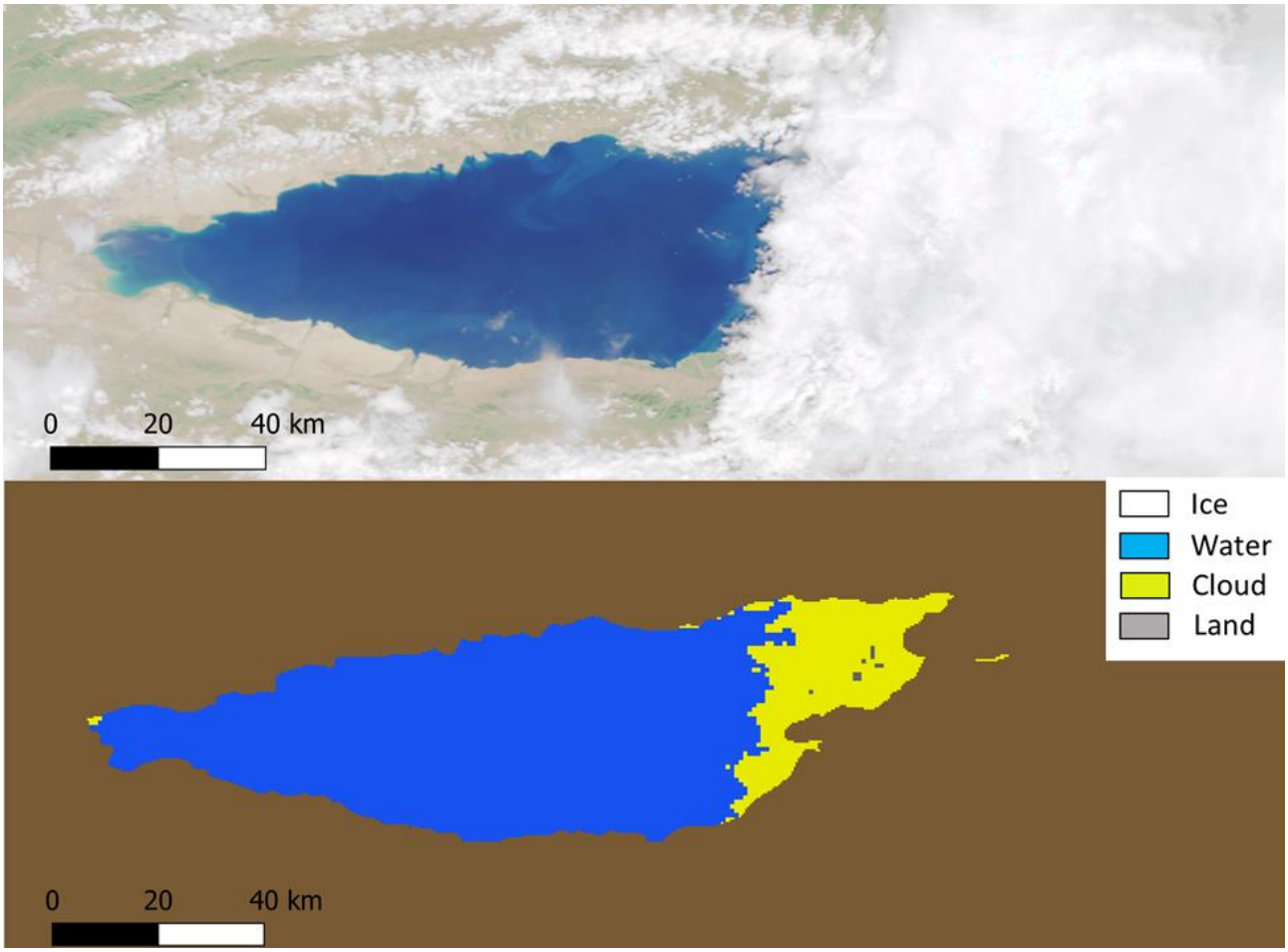


Figure 12. S3 OLCI RGB image and LIE-NH product from Issyk kul, Asia, 11 July 2022. A comparison indicates that LIE-NH cloud detection is not so sensitive to haze over open water, especially when the water is not cold.

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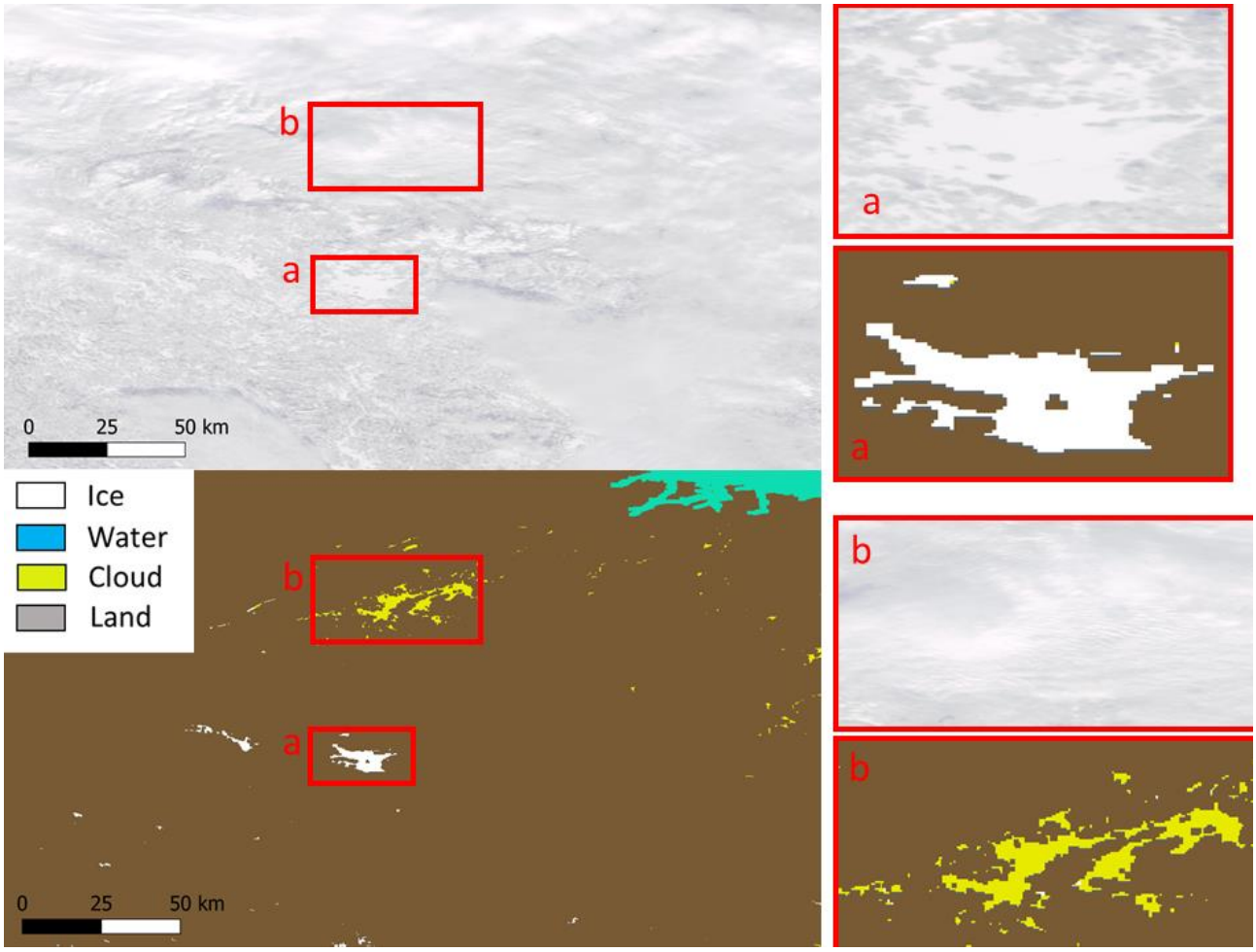


Figure 13. S3 OLCI RGB image and LIE-NH product from Northern Finland, Europe, 12 March 2022. a) Cloud-free area over Lokka reservoir. b) Cloud cover over Lake Inari. A comparison indicates that LIE-NH discriminates well between ice and cloud cover.

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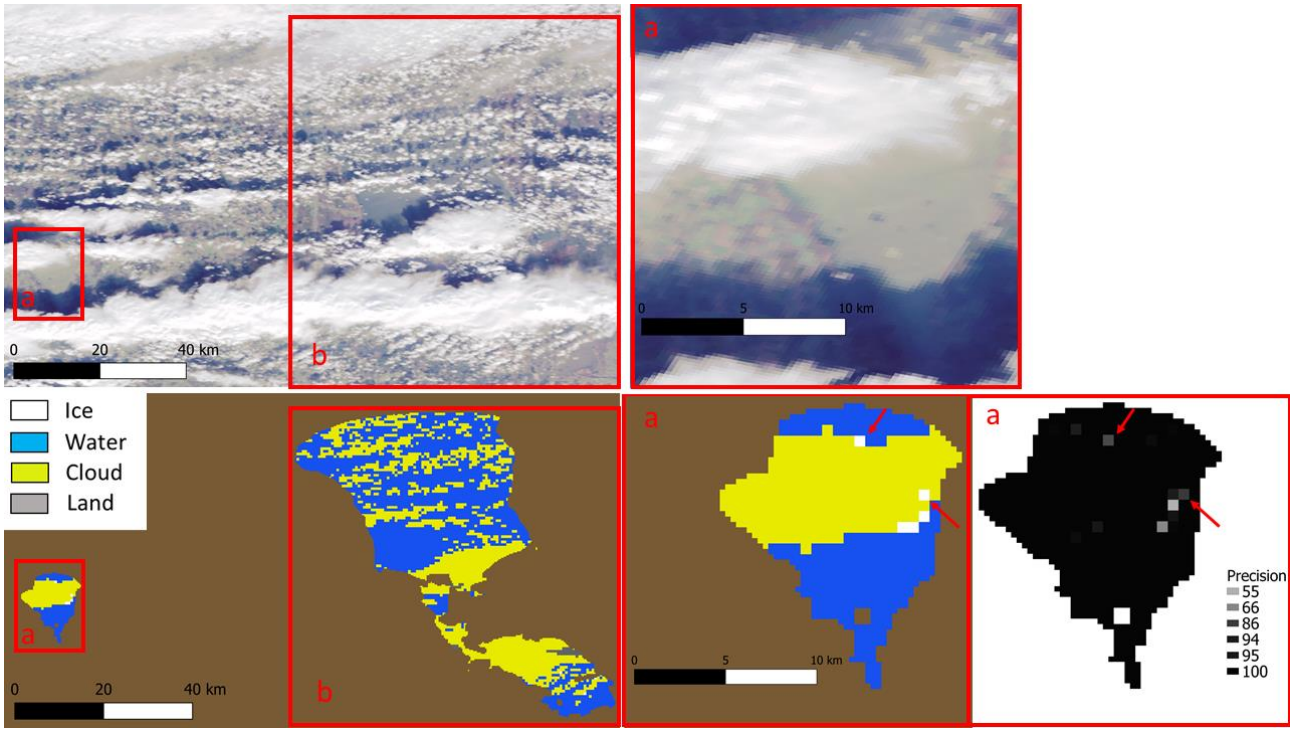


Figure 14. S3 OLCI RGB image and LIE-NH product from Lake Peipus, Europe, 22 October 2021. a) A few pixels at the edge of the cloud over cold and turbid water in Võrtsjärve are incorrectly classified as ice. The precision is also lower at some of the misclassified pixels. b) In lake Peipus the clouds are detected well and there is no misclassified ice even though the light conditions are already poor at the end of October.

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5 CONCLUSIONS

There has not yet been a separate validation for Global LIE. The producers consider that the validation done for LIE-NH is valid as well for Global LIE because the Global LIE utilizes almost the same methodology as LIE-NH at the Northern Hemisphere scale. Furthermore, most seasonally ice-covered lakes are in this region, providing a strong representation of the overall accuracy of the global product. . The validation of LIE-NH was carried out by comparing 20 m high-resolution Lake Ice Extent (LIE) products with the LIE-NH 500 m products, which also use the same *ICEmod* methodology (Heinilä et al. 2021) as the Global LIE. The dataset was spatially and temporally extensive, covering all valid cloud-free data between January 2020 to May 2022 from 45 lakes (15 lakes per each continent; Asia, Europe, North America). The total number of validated cloud-free LIE-NH 500 m resolution lake pixels were over ten million. The 20 m high resolution LIE products were converted to 500 m binary (ice/no-ice) by aggregating in two different ways: 1) the aggregated 500 m pixel is classified as ice-covered if the proportion of 20 m ice pixels is > 50%, 2) the aggregated 500 m pixel is classified as ice-covered if the proportion of 20 m ice pixels is >70%. As a result, 45% of the dataset consists of complex cases (partly ice-covered lakes), 33% of completely ice-covered lakes and 23% completely ice-free lakes. The overall accuracy of LIE-NH product was very good, 99% for both the above-mentioned thresholds (Table 10). The results show that the *ICEmod* algorithm, utilized in the Global LIE product, detects different kinds of ice covers very well. The algorithm is also applicable to distinguish very well turbid cold water from ice. Almost all the inaccuracies were found during the low light conditions in late autumn and mid-winter, and for the complex case lakes. For fully open water lakes, the ice commission error is as low as 0.05%. These rare instances of False Positives occur due to extremely cold and highly turbid water or fog over very cold water. In cases where lakes are fully covered by ice, the ice omission error is 0.5%. False classifications in such cases are primarily caused by the presence of a water layer atop dark ice cover when the ice has begun thawing but remains unbroken. During the freezing and break-up of ice, the product tends to exhibit a higher commission error (3.2%) than omission error (1.7%) for ice. Nevertheless, even during times when the lake is only partly ice-covered, the overall accuracy is as high as over 98%. When the falsely classified ice (FP) cases were investigated more closely, it was evident that ice pieces within the product's 500 m pixel space increase the observed reflectance and cause commission errors. In fragmented ice validation, there are difficulties in deciding whether a 500 m by 500 m pixel in the reference data should be classified as ice-covered or not. The omission errors occur when the method fails to determine whether a case represents *water on dark ice* or *open water*. A more detailed investigation on these omission errors revealed that there were no 500 m reference pixels that had an ice fraction of 100%. In other words, all true ice pixels incorrectly

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classified as water included some 20 m open water pixels. Concerning the end-use of the LIE information, this is likely not crucial, because in those cases, the ice is already weak (citizens), dark (earth energy budget), and partly thawed (ecologist, weather modelers).

Table 10. The binary metrics for LIE-NH product. The 500 m reference pixel is classified as ice if it includes 50% or more 20 m resolution ice pixels.

Metrics	Whole dataset	Reference products representing only open water	Reference products representing only fully ice-covered lakes	Reference products representing lakes which were partly ice-covered, partly ice-free
Number of LIE-NH pixels	10851778	2433541 (23%)	3435519 (32.5%)	47112718 (44.5%)
Recall	98.9%	N/A	99.5%	98.3%
Overall accuracy, Hit rate	99.1%	99.95%	99.5.%	98.3%
Omission error	1.1%	N/A	0.5%	1.7%
Commission error, FAR	0.6%	0.05%	N/A	3.2%
Precision	99.7%	N/A	100%	99.3%
F-score	99.3%	N/A	99.7%	98.8%

The validation of the cloud class was performed by visually comparing dozens of LIE-NH products with S3 OLCI/SLSTR images. Manual validation was chosen, since there is no valid product that detects clouds from S3 SLSTR data over northern lakes well enough to be used as a “truth”. The validation indicated that the *ICEmod* method distinguishes different kind of cloud covers very well from ice and water, but also gave evidence of the algorithm’s sensitivity to detect thin cloud layers especially over ice. The *ICEmod* is not very prone to classify haze or fog over open water as cloud, especially when the water is not cold. This characteristic is advantageous to the areal coverage of the ice information and is due to the utilization of thermal bands in the algorithm. Incorrect classifications (between ice and water) due to the cloud cover were very rarely found, but low light conditions and turbid water increases the possibility of errors at the edge of clouds, where, e.g., fog can be present.

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6 LIMITATIONS AND RECOMMENDATIONS

The Global Lake Ice Extent (Global LIE) is monitored daily using optical satellite data. Observations from optical satellite sensors are limited by cloud cover, as well as low light conditions associated with the polar night from late autumn to early spring in high northern and low southern latitudes. The validation of the LIE-NH product revealed that almost all the inaccuracies occurred during low light conditions in late autumn and mid-winter.

Mapping of ice on lakes requires, especially in Scandinavia and in the North-East of North America, a fine spatial resolution watermask due to the small size of the lakes and relatively high number of islands and narrow parts in some lakes. The gridded Global LIE data product covers the globe (see geographical coverage in section 2 with 0.005° (~500 m) resolution. The shores (also in case of islands) are not included in the product to avoid mixed pixels, i.e., containing both land and water/ice). Sometimes the water body dries out, which can lead to incorrect ice interpretations. This risk is minimized by frequent quality assurance.

The validation results indicate that the thematic overall accuracy requirement of GCOS (AD3) can be achieved. The user is good to be aware of that during the freezing and break-up of ice, the method cannot always detect whether the target represents water on dark (thin, semitransparent) ice or open water. The same goes for the fact that the small white ice floes within a single pixel may result in the pixel being classified as ice. However, in these cases the lake ice is already fragmented and/or dark and weak, and thus does not always cause disadvantages for the end users depending on the use case. The Global LIE product itself does not contain lake morphological, etc. data. If such information is desired, it is recommended that the user combines the product with another water body dataset.

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